

Joseph R. Livesey

CONTACT	Address: 4120 11th Ave NE, Apt 306 Seattle, WA 98105 Phone: +1 (715) 505-3347	Email: jlivesey@uw.edu Website: https://jrlivesey.github.io GitHub: https://github.com/jrlivesey
RESEARCH INTERESTS	Planetary system dynamics, exoplanet atmospheres, planetary formation, astrobiology. Data analysis and modeling, machine learning applications in astronomy.	
EDUCATION	B.S. in Astronomy and Physics , University of Washington <i>Cum laude</i> , Departmental honors in Astronomy and Physics Minors: Mathematics, Linguistics GPA: 3.84	2018–2022 Seattle, WA
RESEARCH POSITIONS	Undergraduate Student Researcher , University of Washington <i>VPlanet Group; Advisor: Rory Barnes</i> Modeling the orbital dynamics and interiors of exoplanets. <i>Condensed Matter Experiment; Advisor: Amal al-Wahish</i> Probing the temperature dependence of proton conduction in alkaline earth phosphates with spectroscopic data.	2019–2022 Seattle, WA
HONORS AND AWARDS	Mary Gates Research Scholarship (\$5,000 award) Quarterly Dean’s List ($\times 11$)	2020 2019–2022
PUBLICATIONS	2. Livesey, J. R. & Barnes, R. 2022, The Secular Orbital Evolution of the Proxima Centauri System , <i>In prep.</i> 1. Barnes, R., Amaral, L. N. R. do, et al. (incl. Livesey, J. R.) 2022, History and Habitability of the SPECULOOS-1 Planetary System , <i>In prep.</i>	
CONFERENCE PROCEEDINGS	2. Barnes, R., Barth, P., et al. (incl. Livesey, J. R.) (2021 March). Interdisciplinary Modeling of Planetary Habitability . Poster presentation at Habitable Worlds 2021, virtual. 1. Livesey, J. R. & Barnes, R. (2021 January). Orbital and Rotational Evolution of Proxima Centauri b . Oral presentation at AAS Meeting 237, virtual.	
CONTRIBUTED TALKS	Dynamics of the Proxima Centauri Planets <i>Undergraduate Research Symposium, UW</i> Orbital and Rotational Evolution of Proxima Centauri b <i>AstroFest, UW</i> Structure and Dynamics Investigation of Newberyite ($\text{MgHPO}_4 \cdot 3\text{H}_2\text{O}$) <i>Undergraduate Research Symposium, UW</i>	2021 <i>Virtual</i> 2020 <i>Virtual</i> 2020 <i>Virtual</i>
SERVICE	Scientific Organizing Committee, VPlanet Workshop	Sept 2022
PROFESSIONAL DEVELOPMENT	VPlanet Developers’ Workshop – <i>Virtual workshop</i>	Jun 2021
ORGANIZATIONS	Phi Beta Kappa UW Mobile Planetarium UW League of Astronomers UW Research Computing Club	2021–Present 2022–Present 2021–2022 2021–2022

SKILLS	Programming: Python, Interactive Python, Mathematica, SQL (<i>proficient</i>) C (<i>familiar</i>)	
	Software: Unix, Git, L ^A T _E X (<i>proficient</i>) HTML, CSS (<i>familiar</i>)	
EMPLOYMENT	UW Department of Astronomy	
	Grader, ASTBIO 115	Spring 2022
	Grader, ASTR 101	Winter 2022
	Odegaard Undergraduate Library	
	Library Technician	2022–Present
	Access Services Student Assistant	2018–2022
REFERENCES	Prof. Rory Barnes , UW Astronomy & Astrobiology	rbk9@uw.edu
	Prof. Eric Agol , UW Astronomy & Astrobiology	agol@uw.edu
	Dr. Amal al-Wahish , UW Physics	wahish@uw.edu
RELEVANT COURSEWORK	ASTRONOMY	PHYSICS
	Solar System/Exoplanets (321, 419) Stellar Astrophysics (322, 421) Extragalactic Astronomy (323) High-Energy Astrophysics (423) Cosmology (425) Programming for Astronomy (300, 302) Astrostatistics & Machine Learning (324) Computational Astrophysics (427) Astronomical Data Analysis (480)	Classical Mechanics (121, 329) Electricity & Magnetism (122, 321, 322) Waves & Optics (123) Thermodynamics (224) Quantum Mechanics (225, 324, 325) Mathematical Methods (227, 228) Particle Physics (226) Statistical Mechanics (328) Atomic Physics (421) Electric Circuits Lab (334) Computational Physics Lab (434) Physics Honors Seminar (485, 486, 487)
	MATHEMATICS	OTHER
	Elementary Calculus (124, 125, 126) Differential Equations (307, 309) Linear Algebra (308) Vector Calculus (324) Probability (394)	Computational Linguistics (LING 471)